

MACHINE LEARNING PROPOSAL

Title: Customer Churn Prediction Using Machine Learning

Problem:

ABC Communications Ltd wants to predict customers who are likely to leave the company before they actually churn. Customer churn can result in loss of revenue and increased customer-acquisition costs.

The proposed machine learning solution will use historical customer information to identify patterns associated with customer churn and predict whether an individual customer is likely to leave the company.

Machine Learning Problem Type

This project is a supervised machine learning classification problem. More specifically, it is a binary classification problem because the target variable, "Churn", has two possible outcomes:

- "Yes" – the customer churned.
- "No" – the customer did not churn.

Target Variable.

The target variable is:

"Churn"

The model will learn from historical customer information and predict the churn status of a customer.

The target contains:

Churn	Number of Customers	Percentage
No	5,174	73.5%
Yes	1,869	26.5%

The target is therefore moderately imbalanced, which should be considered when evaluating model performance.

Input Features:

The dataset contains 21 columns. "customerID" will not be used as a predictive feature because it is an identifier rather than a meaningful customer characteristic. The remaining 19 variables will be used as candidate input features:

- "gender"

- "SeniorCitizen"
- "Partner"
- "Dependents"
- "tenure"
- "PhoneService"
- "MultipleLines"
- "InternetService"
- "OnlineSecurity"
- "OnlineBackup"
- "DeviceProtection"
- "TechSupport"
- "StreamingTV"
- "StreamingMovies"
- "Contract"
- "PaperlessBilling"
- "PaymentMethod"
- "MonthlyCharges"
- "TotalCharges"

Data Preprocessing Plan

The dataset contains both numerical and categorical variables, so appropriate preprocessing will be required before model training.

- **Missing Values**

The inspection identified 11 missing values in "TotalCharges". Examination of these records showed that the affected customers have a tenure of 0 months while their "MonthlyCharges" values are available.

The missing "TotalCharges" values will therefore be treated as 0 during preprocessing.

- Categorical Encoding:

Several input features are categorical and contain values such as "Yes", "No", contract types, internet-service types, and payment methods.

These categorical variables will be converted into numerical representations using One-Hot Encoding where appropriate.

The target variable "Churn" will also need to be converted into a binary numerical representation for model training.

- Numerical Features

The numerical variables include:

- "SeniorCitizen"
- "tenure"
- "MonthlyCharges"
- "TotalCharges"

"SeniorCitizen" is a binary indicator even though it is stored as an integer.

For algorithms that are sensitive to feature scale, such as Logistic Regression, numerical features will be standardized using an appropriate scaling method.

Preprocessing Pipeline

A preprocessing pipeline will be used to ensure that transformations are applied consistently and to reduce the risk of data leakage.

A "ColumnTransformer" can be used to apply different preprocessing operations to numerical and categorical variables, followed by the selected machine learning algorithm in a "Pipeline".

Proposed Machine Learning Algorithms:

Three classification algorithms are proposed for the initial model comparison.

1. Logistic Regression:

Logistic Regression will be used as the baseline classification model.

It is suitable because the problem is binary classification, and the model is relatively simple and interpretable. It can also provide probability estimates for customer churn.

2. Decision Tree Classifier

A Decision Tree will be tested because it can model nonlinear relationships between customer characteristics and churn. It is also relatively easy to interpret and does not require feature scaling.

3. Random Forest Classifier

Random Forest will be evaluated as an ensemble-based alternative to a single Decision Tree.

It combines multiple decision trees and can capture more complex patterns in structured customer data while generally being more robust than an individual decision tree.

Train-Test Strategy

The dataset will be divided into training and testing sets.

The training set will be used to train the models and fit the preprocessing steps, while the testing set will be kept separate for evaluating how well the trained models perform on previously unseen data.

A stratified split will be considered because the target variable is moderately imbalanced. Stratification will help preserve the proportion of churned and non-churned customers in both the training and testing sets.

Model Evaluation Metrics

The proposed models will be evaluated using multiple classification metrics:

- Accuracy

Measures the overall percentage of correct predictions.

- Precision

Measures the proportion of customers predicted as churners who actually churned.

- Recall

Measures the proportion of actual churners that the model successfully identifies.

Recall is particularly important in this project because failing to identify a customer who is likely to churn may result in a missed opportunity for customer retention.

➤ F1-Score

The F1-score combines precision and recall into a single metric and is useful when there is some class imbalance.

➤ ROC-AUC

ROC-AUC will measure the model's ability to distinguish between customers who churn and those who do not across different classification thresholds.

The model comparison will therefore consider accuracy, precision, recall, F1-score, and ROC-AUC, with particular attention given to recall and F1-score.

Expected Outcome:

The expected outcome is a machine learning classification model that can identify customers who are at high risk of churn.

The model should provide useful predictions that ABC Communications Ltd can use to support customer-retention strategies. For example, customers identified as high-risk could be targeted with appropriate retention interventions.

The final model will be selected based on its performance across the chosen evaluation metrics and its suitability for the business objective, rather than simply selecting the model with the highest accuracy.

Proposed Machine Learning Workflow

The proposed workflow is:

Dataset → Data Cleaning → Feature/Target Separation → Train/Test Split → Preprocessing → Encoding & Scaling → Model Training → Model Evaluation → Model Comparison → Final Model Selection

The three proposed models will be trained using a consistent preprocessing strategy and evaluated on the same testing data to ensure a fair comparison.